# HiLabs Product Caselet – APM – Campus Hiring 2026–27

*Reimagining the Case Management Workbench: giving clinicians back their clinical judgment*

**Instructions.** Solve the problem below. Submit **(1) an 8–12 slide deck (or equivalent PDF)** and **(2) a spreadsheet annex** (Excel / Google Sheet) containing synthetic data for **5 patients** that you create. Nobody expects you to know about US healthcare already; researching the domain, finding the real pain points and reasoning with data are part of the test. State your assumptions and cite your sources. We value depth of thinking over breadth of buzzwords.

## 1. Healthcare Primer (5-minute read)

| | |
|---|---|
| **Health plan (payer) & member** | A health plan is the insurer (for example a Medicare Advantage plan). Its customers are called **members**, i.e. the patients. Plans are paid to keep members healthy at a managed cost and are scored on quality (HEDIS measures, CMS Star Ratings, Cost of Care), so they invest in **care management**. |
| **What is a "Case"?** | A goal-driven, time-bound file the plan opens for a member who needs help navigating care, e.g. after a hospital stay, with uncontrolled diabetes, or a high-risk pregnancy. It holds assessments, care plan (goals, barriers, interventions), tasks, every contact attempt, notes and the outcome. It is *not* a claim (a bill) and not a single phone call; it lives for weeks to months. |
| **Who are Case Managers & what do they do?** | Licensed clinicians, mostly RNs plus social workers and behavioral-health counselors, working at the plan, largely by phone or video. They do not treat; they **coordinate and coach**. Typical lifecycle: **identify → outreach → assess** (clinical, behavioral, social needs) **→ plan → coordinate** (appointments, medications, home health, equipment, transport, community resources, authorizations) **→ monitor & follow up → document → close or transition**. |
| **When is a case assigned?** | (1) **Event triggers**: hospital admission, ED visit or discharge alerts (ADT feeds). (2) **Risk models**: high-cost or "rising-risk" scores. (3) **Referrals** from utilization-management nurses, doctors, call center, the member or family. (4) **Program enrollment**: heart failure, COPD, diabetes, kidney disease, oncology, maternity, behavioral health. (5) **Transitions of care**: follow-up after discharge (e.g., contact within ~2 business days, visit within 7). Cases are routed by language, state licensure, program expertise, acuity and the case manager's current capacity. |
| **Typical workload (indicative; varies by plan, validate in your research)** | **Complex case management:** roughly 30–60 active cases per nurse. **Transitions of care / short-term:** 60–100+, with new assignments every day. **Telephonic disease management:** 100–200+. Each case needs repeated contacts over weeks. Documentation (structured assessments, care plans, audit-ready notes required by accreditors and state contracts) competes directly with time spent with members. |

| 1 Identify | 2 Outreach | 3 Assess | 4 Plan | 5 Coordinate & monitor | 6 Close / transition |
|---|---|---|---|---|---|
| Alert, risk score or referral | Reach the member | Clinical, behavioral, social | Goals, barriers, tasks | Care team, services, follow-ups | Outcome documented |

*A case lifecycle (simplified). Every stage is a place where data is needed, and where time is lost.*

## 2. The Problem

**Monday, 8:00 am. Maria is an RN case manager with 20 years of ICU experience and 48 active cases.** Six hospital-discharge alerts arrived since Friday. To understand one member she opens six or seven screens (claims portal, authorization system, pharmacy file, HIE portal, old case notes, her own spreadsheet). The latest claim is 45 days old, the phone number on file is disconnected, two systems disagree on the member's medications, and she must fill a 40-field assessment before closing the contact. By 3 pm she has had two real conversations with patients. Her clinical instinct (noticing that a medication list does not add up, sensing that a caregiver is overwhelmed) is the most valuable thing she brings, and it is barely used.

| Obstacle | What it looks like on the ground |
|---|---|
| **Messy data** | Same member; conflicting diagnoses and medication lists across sources; free-text notes; codes (ICD-10, CPT, NDC, LOINC) with no clinical context; missing or inconsistent fields. |
| **Untimely data** | Claims arrive weeks after care is delivered; hospital alerts show "admitted" but not why or what happened; discharge summaries and labs arrive late, partially, or never. |
| **Work burden** | Unprioritized queues, manual task tracking, side spreadsheets, hunting for "who needs me today?"; caseload spikes without warning. |
| **Documentation** | The same facts re-typed across assessment, care plan, call notes and audit forms; documentation done after the call, often after hours. |
| **Hard outreach** | Wrong or disconnected numbers, voicemail, unknown best time or channel, language and trust barriers, social constraints (no phone minutes, unstable housing, works two jobs). |

**Your challenge.** Design a **Case Management Workbench** MVP (web) for a health plan that: **(a)** assembles a timely, trustworthy, holistic picture of each member; **(b)** tells the case manager whom to work on and what to do next; and **(c)** cuts documentation and outreach friction, so clinicians spend their time on judgment and conversations, not data hunting. Assume a platform like HiLabs can ingest and standardize most healthcare data (claims, clinical records, pharmacy, alerts). **Your job is to decide which data matters, why, and how it should appear on the screen**, not to build pipelines. AI/LLM features are welcome but be explicit about what is automated and what a clinician must confirm.

## 3. What to Submit

| Part | Expected content |
|---|---|
| **A. Understand (1–2 slides)** | Personas (the case manager only); a "day in the life"; your pain points ranked by impact; your estimate of where a case manager's day is lost, with assumptions shown. |
| **B. Data strategy (2 slides)** | Inventory every dataset you would bring into the product (as many as you can justify, e.g., think beyond claims). For each: **source, what it tells the clinician, how fresh it is, expected quality problems and how you would handle them, and priority** (must-have vs. later). Show how datasets combine into insights (e.g., discharge alert + pharmacy fills + appointment data = "has not picked up heart-failure medication since discharge"). |
| **C. Synthetic patients (annex + 1–2 slides)** | **Create 5 fictional patients**, each stressing a *different* scenario (ideas, not a checklist: fresh heart-failure discharge; rising-risk diabetic who stopped refills; unreachable frequent ED user; behavioral health plus housing instability; high-risk pregnancy or a dual-eligible senior with caregiver strain; or your own). For each patient populate sample records across **at least 6 of your datasets** and plant a few realistic imperfections (duplicate ID, conflicting medication lists, delayed claim, wrong phone). Add one line per patient: "what should the case manager see and do at 8 am Monday?" Use clearly fake data; **no real PHI**. |
| **D. Product design (3–4 slides)** | Wireframes of **at least 3 key screens** (e.g., prioritized work queue, member 360 view, outreach and documentation view; your choice). Explain what sits where and why: what is above the fold, how data freshness, conflicts and confidence are shown, how "next best action" works, and where AI assists. **Walk 2–3 of your synthetic patients through the wire frames.** |

| Part | Expected content |
|---|---|
| **E. Prioritize & measure (1 slide)** | MVP vs. later, and your prioritization logic; success metrics with baseline and target (e.g., prep time per case, active cases per manager, outreach connect rate etc.); risks (wrong or stale data, alert fatigue, clinician trust) and mitigations. |

**Brownie points:** a pilot and go-to-market approach for a first health plan (buyer, pilot scope, pricing signal).

**Format example for the synthetic-data annex (illustrative; do not reuse this patient) – there are many more data sources which are essential and helpful in creating a solution**

| Dataset | Sample record for "Patient X – 74-year-old, Medicare Advantage, COPD + diabetes" | Planted imperfection |
|---|---|---|
| **Member & eligibility** | ID M-10482; plan MA-PPO; PCP Dr. A. Rao; language: Spanish | No Imperfection – assume that the health plans have unique patient identifiers across datasets |
| **Hospital alert (ADT)** | ED visit 28 Oct 2026, 09:40; no diagnosis in alert | Arrived 2 days later; discharge status unknown |
| **Claims** | Last claim: inhaler refill billed 12 Sep; | Claims lag of 30+ days hides the ED visit |
| **Pharmacy fills** | Tiotropium last filled 3 Aug (90-day supply); metformin filled on time | COPD inhaler gap of ~60 days; not visible on the claims screen |
| **Contact history** | 4 outreach attempts in 3 weeks, all voicemail; phone type "home" | Number belongs to a daughter; best time unknown |
| **SDOH screen** | Lives alone; transport "sometimes a problem"; food insecure | Self-reported 8 months ago, possibly stale |

*Expected "8 am Monday" line:* recent ED visit, likely COPD flare after a ~60-day inhaler gap; call the daughter's number in the evening, in Spanish; arrange a pharmacy refill and a ride to the PCP.

### Ground rules & tips for a standout submission

- **Public information and your own thinking only.** AI tools are fine for research, but you must be able to defend every number and design choice.
- **Tie every dataset and feature to a decision.** "What would Maria do differently because she saw this?" is the test we will apply to every item on your slides.
- **Show your numbers.** If you claim documentation takes a third of a case manager's day, show how you got there and how sure you are.
- **Be opinionated.** Tell us what you would build first, what you would deliberately leave out.

## 4. How We Will Evaluate

| Dimension | What great looks like | Weight |
|---|---|---|
| **Problem depth & research** | Real pain points grounded in sources and your own estimates; a clear view of who the user is and what a bad day costs them. | 20% |
| **Data thinking** | A rich, *justified* dataset list; each one tied to a decision it enables smart combinations. Depth beats a long list of names. | 25% |
| **Synthetic data & scenarios** | Five distinct, realistic patients with believable imperfections that genuinely stress the product design. | 20% |
| **Product design & prioritization** | Information hierarchy that follows the clinician's workflow; thoughtful use of AI with a human in the loop; a defensible MVP. | 25% |
| **Metrics, risks & communication** | Measurable outcomes, sensible risks, and a crisp storyline a busy executive can follow in minutes. | 10% |

## 5. Quick Reference & Suggested Reading

**Terms:** *ADT* (admit/discharge/transfer alerts) · *EHR / CCD* (clinical records) · *HIE* (health information exchange) · *UM* (utilization management, i.e., prior authorizations) · *HRA* (health risk assessment) · *SDOH* (social determinants of health, e.g., housing, food, transport) · *HEDIS* (quality measures) · *ICD-10 / CPT / NDC / LOINC* (diagnosis, procedure, drug and lab codes).